Geosteering Under Structural Uncertainty

Project: GeoSteering Structural Uncertainty
Synthetic causal run and simplified causal comparison. Training and portfolio tool; not validated for live operations.
Run ID: e928f8d1-2407-428e-adf0-413bedc332c1
Status: COMPLETE; revealed MD (ft): 7000.0

Measured result

Over the same 5000-ft lateral, HOLD placed 38.07% (1903.6 ft) in zone; the observation-guided causal policy placed 97.17% (4858.7 ft) in zone. Lateral DLS violation counts: HOLD 0, guided 0. This controlled synthetic result relies on the disclosed signed layer calibration. Replay recommendations did not change the recorded path.
scenario_version: structural_dip_01-v1
seed: 123
code_revision: 9176b26a74972cdcaa8d0b41d1737bfa2c3fe8a3
dirty_working_tree: True
coordinate_reference: LOCAL_EAST_NORTH_FT; shared declared vertical datum; TVD positive down

Data summary and QC

Well	Curves	QC valid	Warnings
active_01	GR, RES	True	
offset_01	GR, RES	True	
offset_02	GR, RES	True	
offset_03	GR, RES	True	

Input provenance (SHA-256)

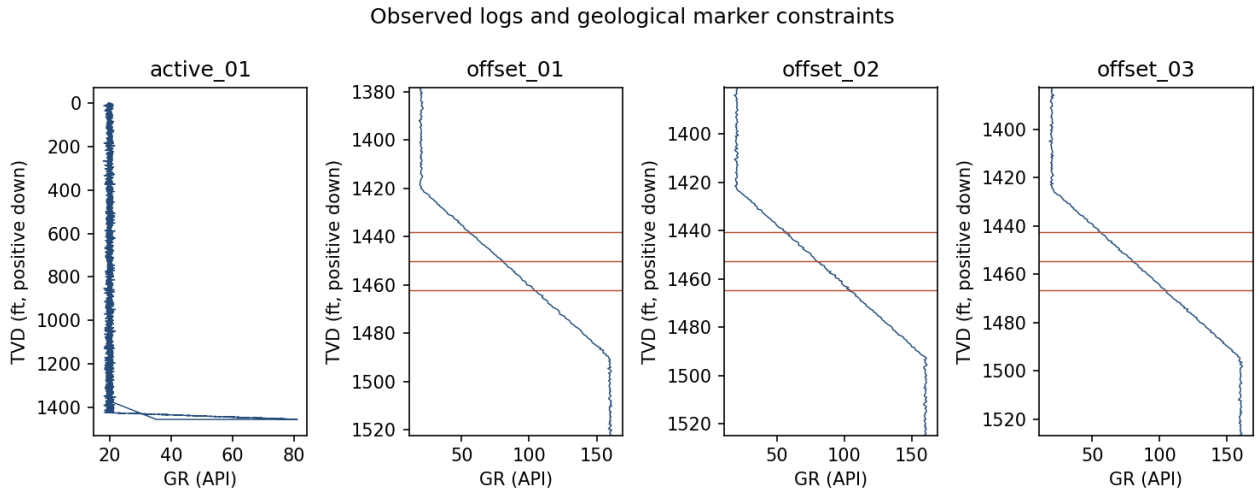
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Correlation and markers

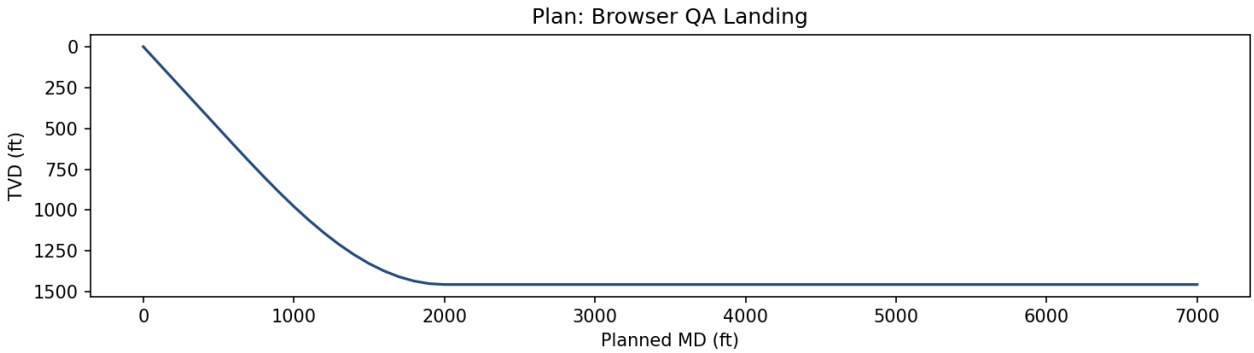
Offset formation picks constrain a regional estimate in shared along-track x and local TVD. Active observations and markers are limited to the saved revealed depth.

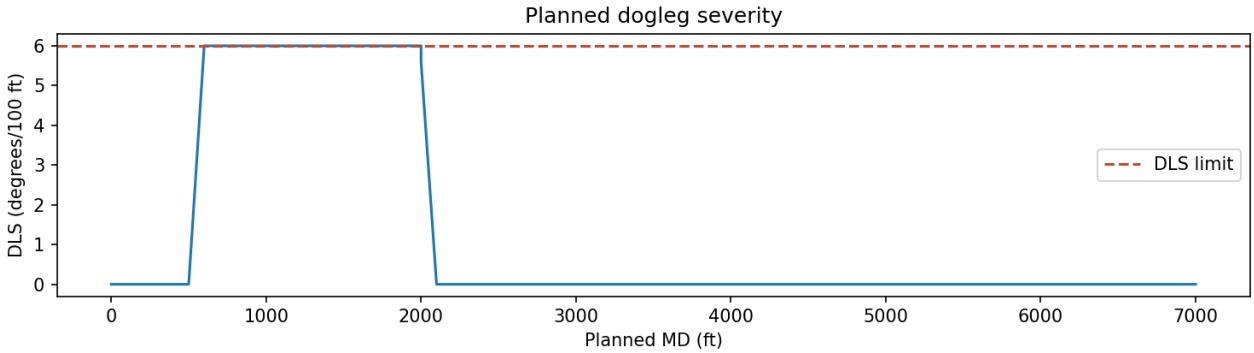
Well	Marker	MD (ft)
offset_01	Top_A	1438.16
offset_01	Mid_A	1450.16
offset_01	Base_A	1462.16
offset_02	Top_A	1440.66
offset_02	Mid_A	1452.66
offset_02	Base_A	1464.66
offset_03	Top_A	1442.66
offset_03	Mid_A	1454.66
offset_03	Base_A	1466.66



Pre-spud plan and DLS

Plan name: Browser QA Landing; Plan ID: 9165372a-7057-45c1-a2f6-6b968d30dfca
Max DLS constraint: 6.0 degrees/100 ft; exceedance count: 0



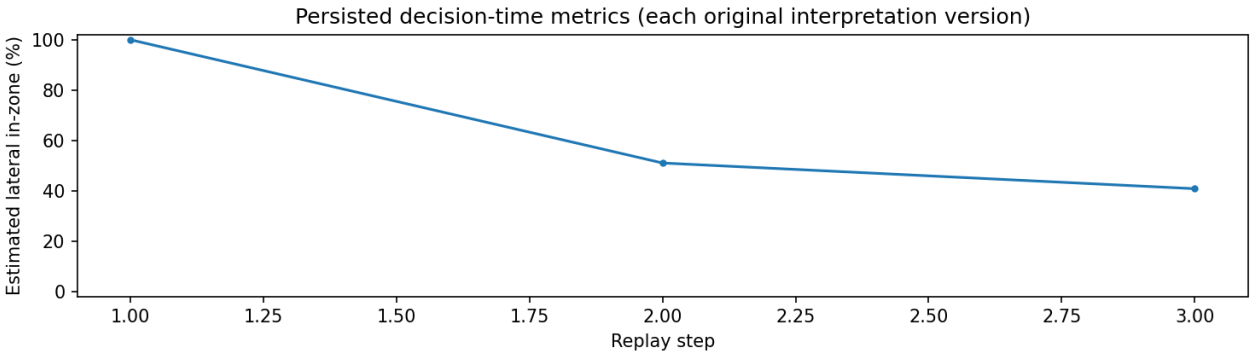


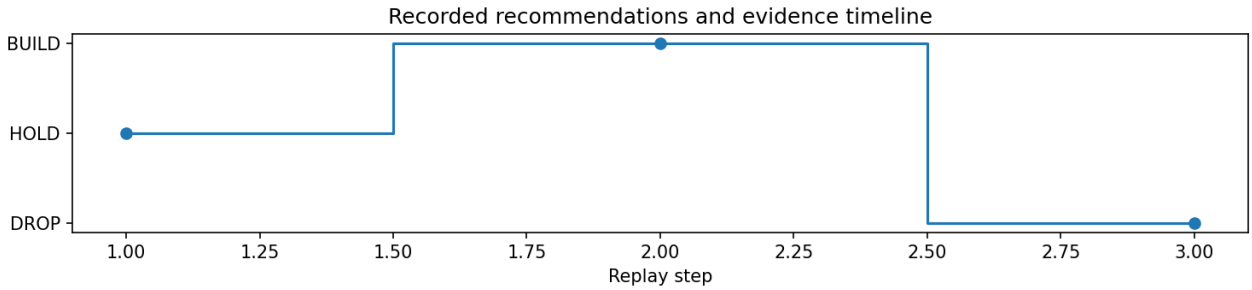
Steering decisions and KPI results

Applied commands change the acquired synthetic trajectory under bounded DLS. Estimated metrics are frozen at each reveal using the cited interpretation, and exclude the build/landing interval.

Step	MD interval (ft)	Action	Evidence / interpretation version
1	2000-4000	HOLD	Acquired landing measurements recorded. Manual HOLD exercises this synthetic causal interface.; version cdfb3916-5bb3-407b-8203-c559c581d70e
2	4000-6000	BUILD	Acquired survey/log prefix ends at MD4000 ft. Controlled BUILD requests actuator clipping in this synthetic UI exercise, not a field recommendation.; version 26a9fb83-9045-4c6e-8c4d-a22d9692bff4
3	6000-7000	DROP	Acquired survey/log prefix ends at MD6000 ft. Controlled DROP exercises the opposite bounded actuator response; no operational safety claim.; version 2f4b79b2-a6c7-4627-ba41-e08cfecec92c

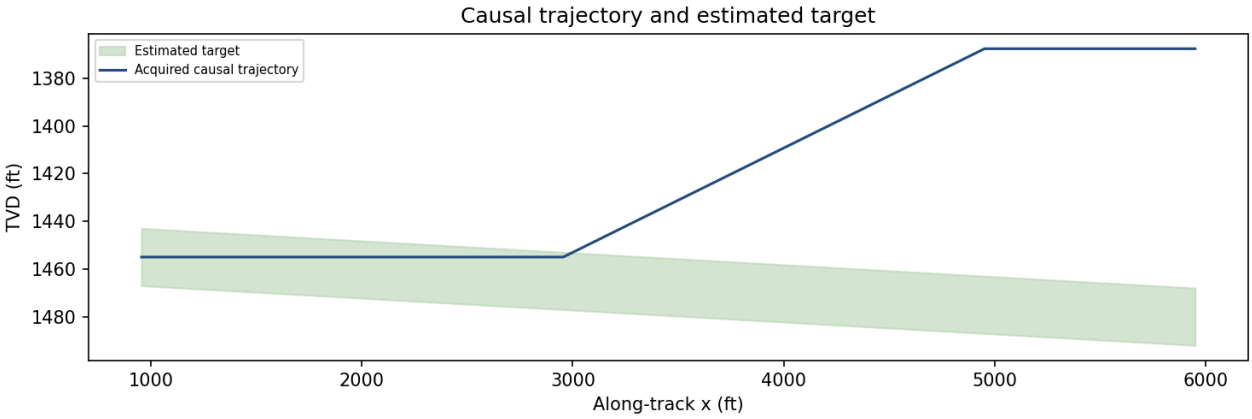
Step	Revealed MD (ft)	Estimated in-zone (%)	Mean center distance (ft)
1	4000	100.000	5.000
2	6000	51.029	31.800
3	7000	40.823	47.379





Final interpretation

Interpretation version: 2f4b79b2-a6c7-4627-ba41-e08cfecec92c; rationale: Carry forward current interpretation: Acquired survey/log prefix ends at MD6000 ft. Controlled DROP exercises the opposite bounded actuator response; no operational safety claim.
Assumed uncertainty band (ft): 10.0. This scenario-assumed range is not a calibrated statistical confidence interval.

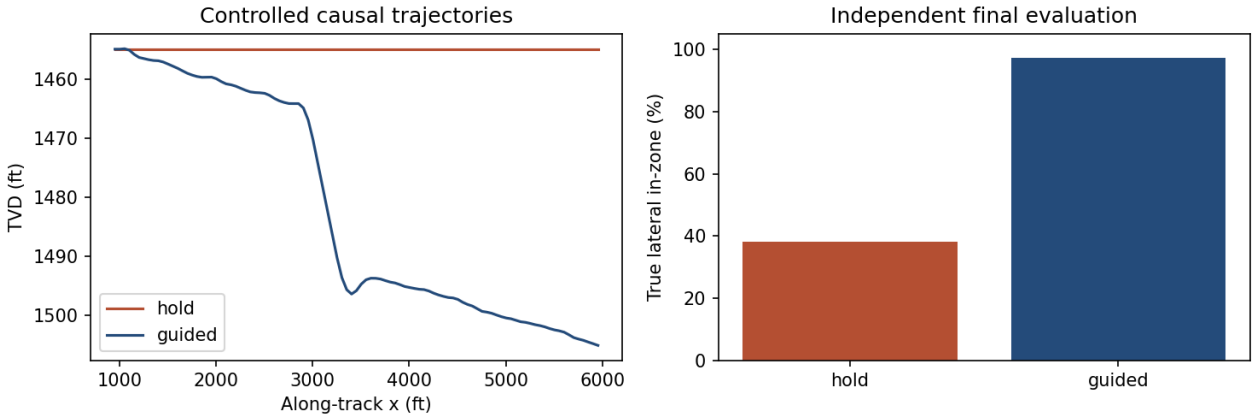


Independent evaluation

Withheld geological truth was used only after the causal run completed. The acquired path reflects applied bounded commands.
interpretation_error: {'baseline_interpretation_mean_absolute_error_ft': 15.778996367571722,
'final_interpretation_mean_absolute_error_ft': 15.778996367571722}
lateral_start_md_ft: 2000.0
limitation: Simplified kinematics and synthetic sensors; not operational drilling software
metric_basis: withheld_truth
metrics: {'avg_center_distance_ft': 62.403213780791, 'dls_violation_count': 0, 'evaluated_length_ft': 5000.0, 'in_zone_length_ft': 1903.5810398298559, 'integration_step_max_ft': 1.0, 'max_dls_deg_per_100ft': 1.0000000000000004, 'max_overshoot_ft': 125.19206430573013, 'percent_in_zone': 38.07162079659712}

mode: causal
run_id: e928f8d1-2407-428e-adf0-413bedc332c1
scenario_version: structural_dip_01-v1
seed: 123
truth_sha256: daa41e8f1685302418efb93bcc7d82c78c37fd6fdc2868fbfc57cd6ca50edbd3

Causal strategy	True in-zone (%)	In-zone / evaluated length (ft)	Maximum lateral DLS
guided	97.174	4858.7 / 5000.0	3.0000
hold	38.072	1903.6 / 5000.0	0.0000



Assumptions and limitations

Synthetic calibration: GR = 80 + 2*(TVD - target center) API, clipped to 20-160 API, with 0.6 API Gaussian noise. This deliberately signed layer signal is known to the synthetic operator; real gamma-ray inversion does not generally provide this information. Saturation cannot identify an exact center and is logged as unresolved evidence.
Scoring: estimated zones use exact piecewise-linear MD interval integration. Nonlinear withheld geology uses at most 1-ft intervals with boundary-crossing interpolation; reference refinement to 0.1 ft changes in-zone results by less than 0.001 percentage points. The final-point indicator is not an interval-length percentage.
Coordinates: feet; MD along the well, local TVD positive down, along-track x eastward. Minimum-curvature integration uses survey inclination and azimuth. Layer metrics are length weighted over the lateral interval with an explicit evaluated-length denominator.
The operator baseline uses offset picks only. Subsequent interpretations may use only revealed active measurements. Geological truth remains fixed and is isolated from live interpretations and recommendations.
Causal trials share geology, initial state, seed and trajectory constraints. HOLD/BUILD/DROP use a simplified bounded kinematic response. Report measured outcomes, including failure; no production drilling, sensor physics or operational safety claims are made.
Idealized 2D geology and synthetic GR/resistivity response; sparse offsets and assumed uncertainty. No real-time telemetry or WITSML support. Numerical summaries are reproducible; PDF timestamps and run identifiers can differ.